

ENTERPRISE NETWORK SIMULATION PROJECT

Project Report

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Academic Year: 3rd Year

Date: 2026

1. Executive Summary

This report documents the design, implementation, and verification of a fully functional enterprise network simulation built using Cisco Packet Tracer. The project was undertaken as part of my third year Bachelor of Information and Communication Technology studies at the University of Kelaniya, Faculty of Computing and Technology, with a specialization in Computer Network Technology.

The simulation replicates a real-world enterprise network consisting of two geographically separated office locations — a Headquarters (HQ) site and a Branch site — connected via a Wide Area Network (WAN) serial link. The network incorporates industry-standard technologies and protocols including VLAN segmentation, inter-VLAN routing, OSPF dynamic routing, DHCP, DNS, Access Control Lists (ACLs), port security, and SSH remote management.

The primary objective of this project was to gain practical, hands-on experience with the technologies used in real enterprise network environments, and to demonstrate competency in network design, configuration, and security — skills directly applicable to a network engineer or network security internship role.

KEY ACHIEVEMENT: All configurations were successfully implemented and verified. End-to-end connectivity between HQ and Branch was confirmed, DHCP and DNS services were validated, security policies were enforced via ACLs and port security, and secure remote management via SSH was confirmed on all network devices.

2. Network Topology

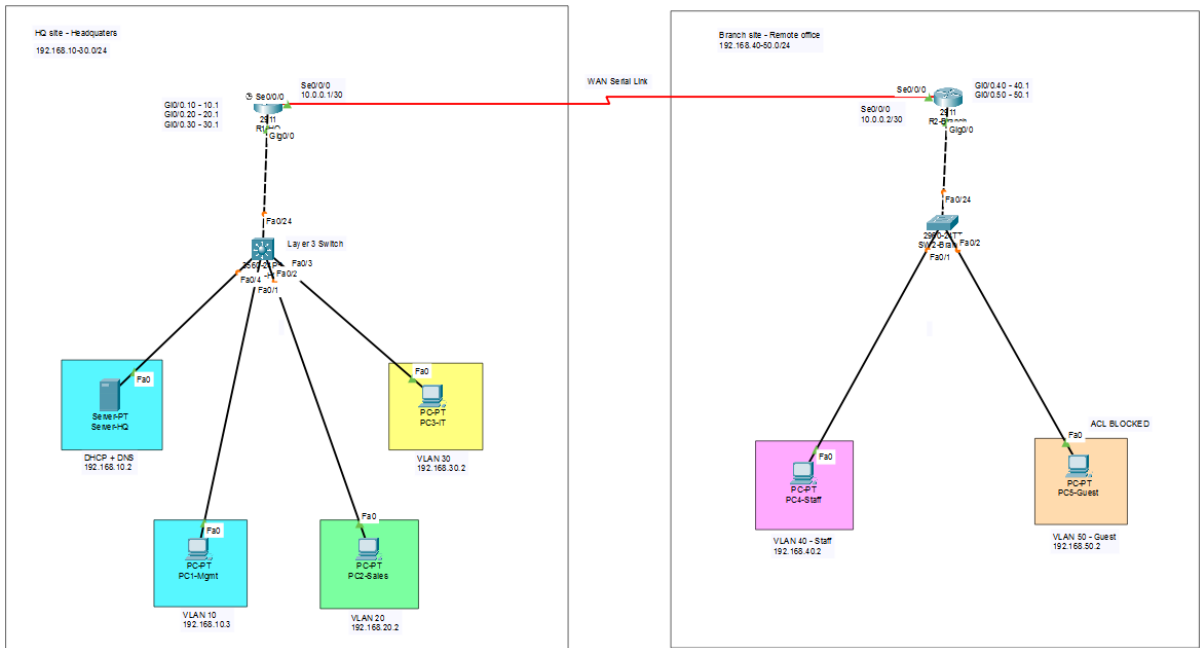
2.1 Topology Overview

The network consists of two sites connected via a serial WAN link. The topology was designed to simulate a typical small-to-medium enterprise environment with departmental VLAN segmentation and centralized services at the HQ site.

Component	Details
HQ Site	Cisco 2911 Router (R1-HQ), Cisco 3560 Layer 3 Switch (SW1-HQ), DHCP/DNS Server, 3 PCs
Branch Site	Cisco 2911 Router (R2-Branch), Cisco 2960 Layer 2 Switch (SW2-Branch), 2 PCs
WAN Link	Serial DCE cable between R1-HQ and R2-Branch using 10.0.0.0/30 subnet
VLANs	5 VLANs — VLAN 10 Management, 20 Sales, 30 IT, 40 Staff, 50 Guest
Routing Protocol	OSPF Area 0 — dynamic routing between HQ and Branch
Services	DHCP (auto IP assignment), DNS (hostname resolution)
Security	ACLs (Guest VLAN isolation), Port Security, SSH (encrypted management)

2.2 Topology Diagram

The topology diagram below illustrates the complete network architecture including both sites, device connections, VLAN assignments, and the WAN link.



HQ site — Headquarters

192.168.10-30.0/24

Gi0/0.10 → .10.1
Gi0/0.20 → .20.1
Gi0/0.30 → .30.1

R1-HQ

Cisco 2911 — Router-on-a-Stick

Se0/0/0
10.0.0.1/30

Trunk Fa0/24

SW1-HQ

Cisco 3560 — L3 switch

Fa0/4

Fa0/1

Fa0/2

Fa0/3

Server-HQ

DHCP + DNS
192.168.10.2

PC1-Mgmt

VLAN 10
192.168.10.x

PC2-Sales

VLAN 20
192.168.20.x

PC3-IT

VLAN 30
192.168.30.x

OSPF Area 0
10.0.0.1 — 10.0.0.2

WAN
Serial link

Branch site — Remote office

192.168.40-50.0/24

Gi0/0.40 → .40.1
Gi0/0.50 → .50.1

R2-Branch

Cisco 2911 — Router

Se0/0/0
10.0.0.2/30

Trunk Fa0/24

SW2-Branch

Cisco 2960 — L2 switch

Fa0/1

Fa0/2

PC4-Staff

VLAN 40 — Staff
192.168.40.x

PC5-Guest

VLAN 50 — Guest
192.168.50.x

Legend

- HQ devices
- Branch devices
- Guest (blocked)
- - - WAN / ACL
- LAN links

ACL blocked

3. IP Address Scheme

3.1 Complete IP Address Table

The following table documents the complete IP addressing scheme used across the enterprise network. A structured addressing plan was created before implementation to ensure logical segmentation and scalability.

Device	Interface/VLAN	IP Address	Subnet Mask	Default Gateway
R1-HQ	GigabitEthernet0/0.10	192.168.10.1	255.255.255.0	—
R1-HQ	GigabitEthernet0/0.20	192.168.20.1	255.255.255.0	—
R1-HQ	GigabitEthernet0/0.30	192.168.30.1	255.255.255.0	—
R1-HQ	Serial0/0/0	10.0.0.1	255.255.255.252	—
R2-Branch	GigabitEthernet0/0.40	192.168.40.1	255.255.255.0	—
R2-Branch	GigabitEthernet0/0.50	192.168.50.1	255.255.255.0	—
R2-Branch	Serial0/0/0	10.0.0.2	255.255.255.252	—
Server-HQ	VLAN 10	192.168.10.2	255.255.255.0	192.168.10.1
PC1-Mgmt (HQ)	VLAN 10	DHCP — 192.168.10.3	255.255.255.0	192.168.10.1
PC2-Sales (HQ)	VLAN 20	DHCP — 192.168.20.2	255.255.255.0	192.168.20.1
PC3-IT (HQ)	VLAN 30	DHCP — 192.168.30.2	255.255.255.0	192.168.30.1
PC4-Staff (Branch)	VLAN 40	DHCP — 192.168.40.2	255.255.255.0	192.168.40.1
PC5-Guest (Branch)	VLAN 50	DHCP — 192.168.50.2	255.255.255.0	192.168.50.1

3.2 VLAN Design

Five VLANs were designed to logically segment the network by department and access level. This segmentation improves security, reduces broadcast traffic, and mirrors real enterprise network design practices.

VLAN ID	Name	Site	Subnet	Purpose
10	Management	HQ	192.168.10.0/24	Network management and servers
20	Sales	HQ	192.168.20.0/24	Sales department users
30	IT	HQ	192.168.30.0/24	IT department users
40	Staff	Branch	192.168.40.0/24	Branch office staff
50	Guest	Branch	192.168.50.0/24	Guest users — restricted access

4. Network Configuration

4.1 VLAN Configuration

VLANs were created and assigned to switch ports on both SW1-HQ (Layer 3 switch) and SW2-Branch (Layer 2 switch). Trunk ports were configured to carry all VLAN traffic between switches and routers.

SW1-HQ VLAN configuration:

```
vlan 10
 name Management
vlan 20
 name Sales
vlan 30
 name IT
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
```

NOTE: The Cisco 3560 Layer 3 switch requires 'switchport trunk encapsulation dot1q' before 'switchport mode trunk'. This is not required on the 2960 Layer 2 switch.

4.2 Inter-VLAN Routing — Router-on-a-Stick

Inter-VLAN routing was implemented on R1-HQ using the Router-on-a-Stick technique. One physical Gigabit interface was divided into three logical sub-interfaces, each handling traffic for one VLAN. The same approach was used on R2-Branch for Branch VLANs.

```
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
interface GigabitEthernet0/0
no shutdown
```

```
R1-HQ
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1-HQ
R1-HQ(config)#
R1-HQ(config)#interface GigabitEthernet0/0.10
R1-HQ(config-subif)#encapsulation dot1Q 10
R1-HQ(config-subif)#ip address 192.168.10.1 255.255.255.0
R1-HQ(config-subif)#exit
R1-HQ(config)#
R1-HQ(config)#interface GigabitEthernet0/0.20
R1-HQ(config-subif)#encapsulation dot1Q 20
R1-HQ(config-subif)#ip address 192.168.20.1 255.255.255.0
R1-HQ(config-subif)#exit
R1-HQ(config)#
R1-HQ(config)#interface GigabitEthernet0/0.30
R1-HQ(config-subif)#encapsulation dot1Q 30
R1-HQ(config-subif)#ip address 192.168.30.1 255.255.255.0
R1-HQ(config-subif)#exit
R1-HQ(config)#
R1-HQ(config)#interface GigabitEthernet0/0
R1-HQ(config-if)#no shutdown
R1-HQ(config-if)#
%LINE-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINE-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINE-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINE-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
R1-HQ(config-if)#end
R1-HQ#
%SYS-5-CONFIG_1: Configured from console by console
R1-HQ#write memory
% Invalid input detected at '' marker.
R1-HQ#write memory
Building configuration...
[OK]
R1-HQ#
```


4.3 WAN Link and OSPF Dynamic Routing

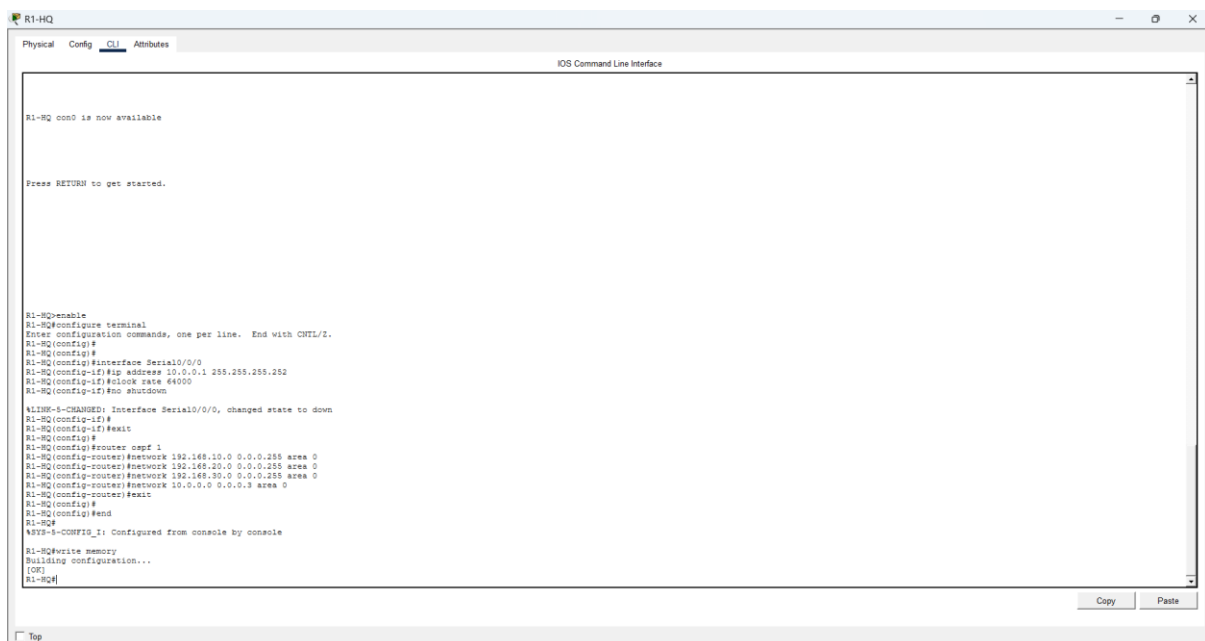
A serial WAN link was established between R1-HQ and R2-Branch using a point-to-point /30 subnet (10.0.0.0/30). OSPF Area 0 was configured on both routers to enable dynamic route exchange between the two sites.

R1-HQ Serial interface:

```
interface Serial0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
```

R1-HQ OSPF configuration:

```
router ospf 1
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```



```
R1-HQ
Physical Config CLI Attributes
IOS Command Line Interface

R1-HQ con0 is now available

Press RETURN to get started.

R1-HQenable
R1-HQ#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1-HQ(config)#
R1-HQ(config)#interface Serial0/0/0
R1-HQ(config-if)#ip address 10.0.0.1 255.255.255.252
R1-HQ(config-if)#clock rate 64000
R1-HQ(config-if)#no shutdown
%LINK-3-CHANGED: Interface Serial0/0/0, changed state to down
R1-HQ(config-if)#
R1-HQ(config-if)#exit
R1-HQ(config)#
R1-HQ(config)#router ospf 1
R1-HQ(config-router)#network 192.168.10.0 0.0.0.255 area 0
R1-HQ(config-router)#network 192.168.20.0 0.0.0.255 area 0
R1-HQ(config-router)#network 192.168.30.0 0.0.0.255 area 0
R1-HQ(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1-HQ(config-router)#exit
R1-HQ(config)#
R1-HQ(config)#end
R1-HQ#
%SYS-5-CONFIG_I: Configured from console by console

R1-HQ#write memory
Building configuration...
[OK]
R1-HQ#
```

4.4 DHCP Server Configuration

The Server-HQ was configured as a centralized DHCP server with separate address pools for each of the five VLANs. The ip helper-address command was applied on all router sub-interfaces to forward DHCP broadcast requests from each VLAN to the server.

Server-HQ

Physical Config **Services** Desktop Programming Attributes

SERVICES

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: VLAN50

Default Gateway: 192.168.50.1

DNS Server: 192.168.10.2

Start IP Address: 192.168.50.168

Subnet Mask: 255.255.255.255

Maximum Number of Users: 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add **Save** **Remove**

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN50	192.168.50.1	192.168.10.2	192.168.50.0	255.255.255.0	256	0.0.0.0	0.0.0.0
VLAN40	192.168.40.1	192.168.10.2	192.168.40.0	255.255.255.0	256	0.0.0.0	0.0.0.0
VLAN30	192.168.30.1	192.168.10.2	192.168.30.0	255.255.255.0	256	0.0.0.0	0.0.0.0
VLAN20	192.168.20.1	192.168.10.2	192.168.20.0	255.255.255.0	256	0.0.0.0	0.0.0.0
VLAN10	192.168.10.1	192.168.10.2	192.168.10.0	255.255.255.0	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Top

4.5 DNS Server Configuration

The Server-HQ was also configured as a DNS server. Hostname records were created to allow devices to resolve domain names instead of IP addresses, simulating a real enterprise internal DNS setup.

Server-HQ

Physical Config **Services** Desktop Programming Attributes

SERVICES

DNS Service: ☒ On ☐ Off

Resource Records

Name: _____ Type: [ARecord]

Address: _____

Add **Save** **Remove**

No.	Name	Type	Detail
0	branch.company.local	ARecord	192.168.40.1
1	hq.company.local	ARecord	192.168.10.2
2	server.company.local	ARecord	192.168.10.2

DNS Cache

Top

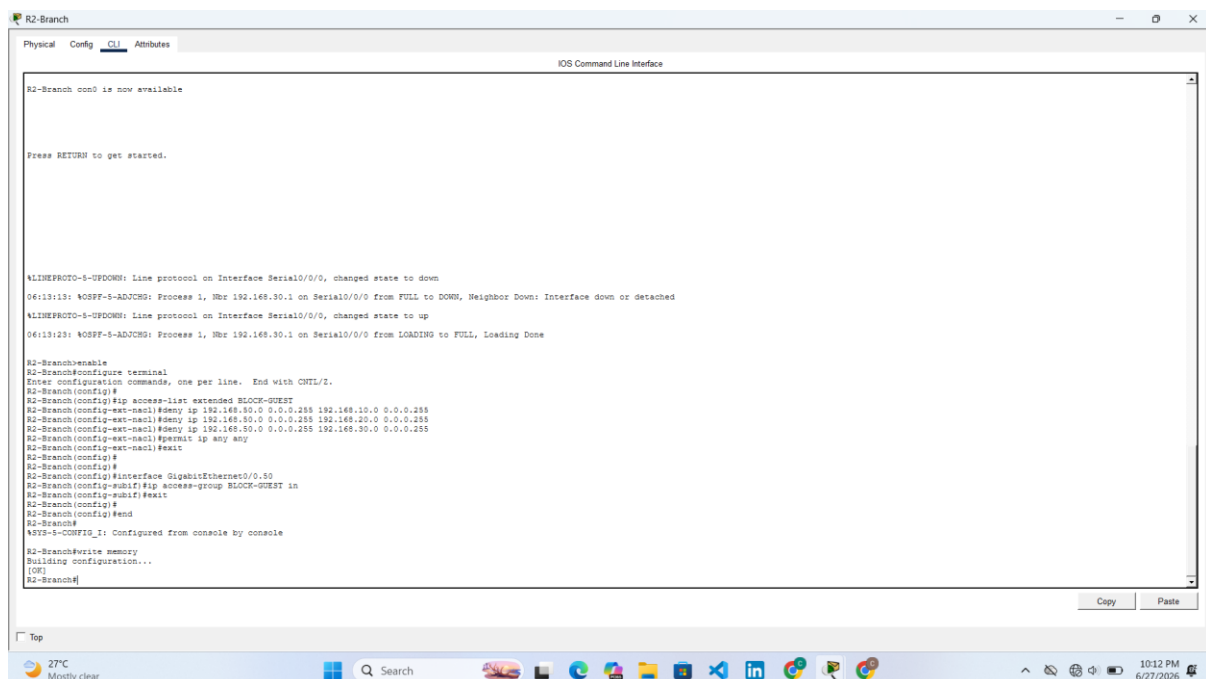
5. Security Implementation

5.1 Access Control Lists (ACLs)

An extended named ACL named BLOCK-GUEST was configured on R2-Branch to enforce network security policy. The Guest VLAN (VLAN 50) was completely blocked from accessing any HQ site resources including Management, Sales, and IT subnets. This mirrors a real guest WiFi policy where visitors can access the internet but cannot reach internal corporate resources.

```
ip access-list extended BLOCK-GUEST
deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
permit ip any any
```

```
interface GigabitEthernet0/0.50
ip access-group BLOCK-GUEST in
```

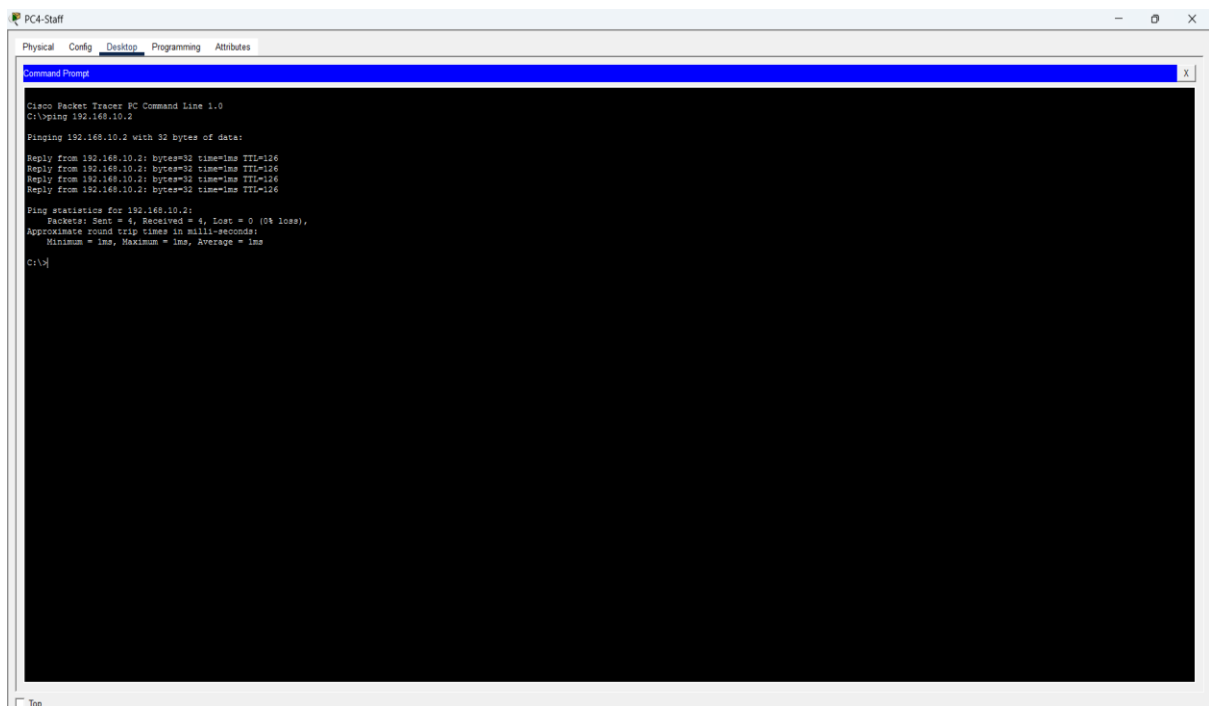
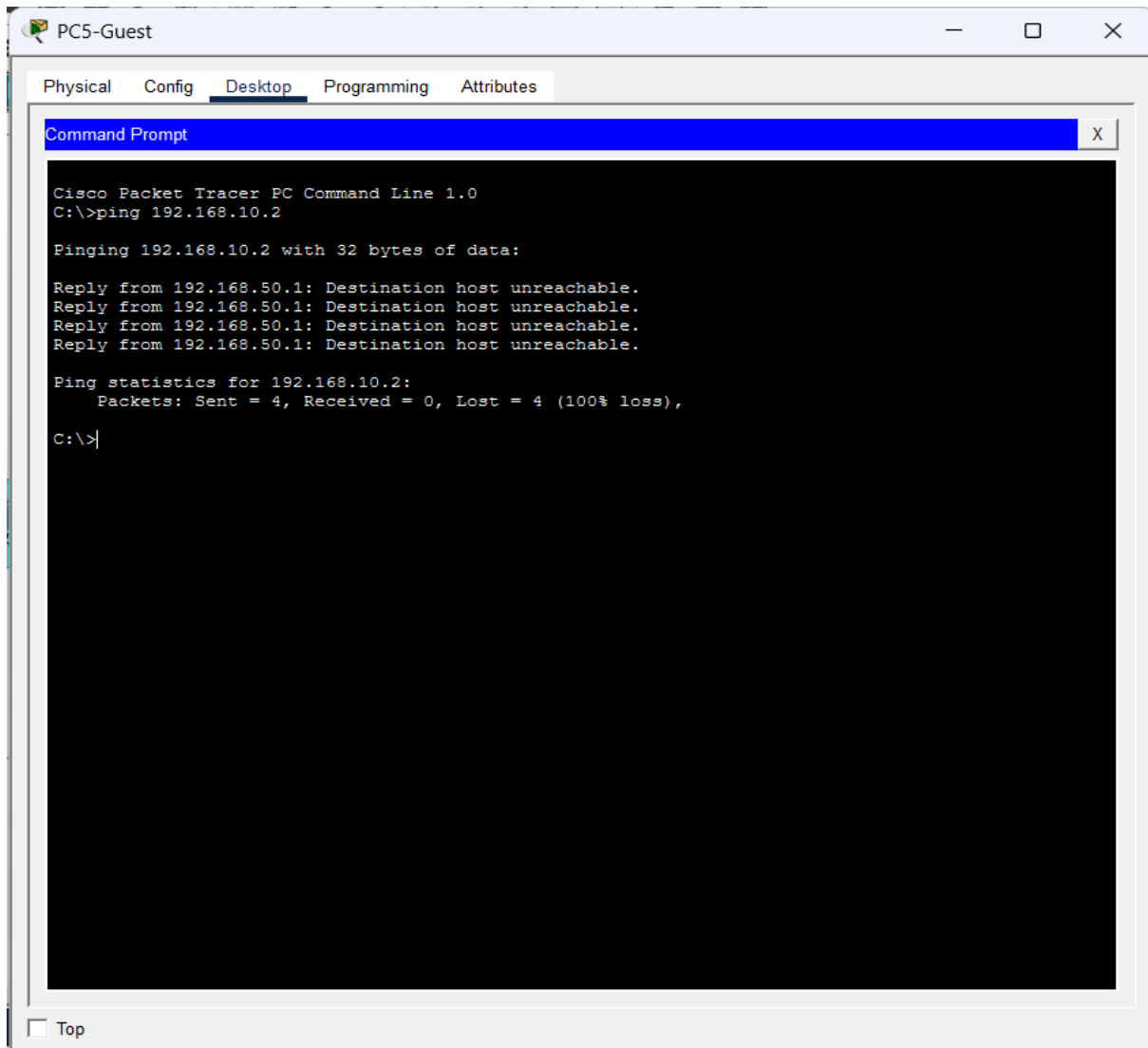


```
R2-Branch
R2-Branch com0 is now available

Press RETURN to get started.

VLINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down
06:13:13: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
VLINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
06:13:13: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on Serial0/0/0 from LOADING to FULL, Loading Done

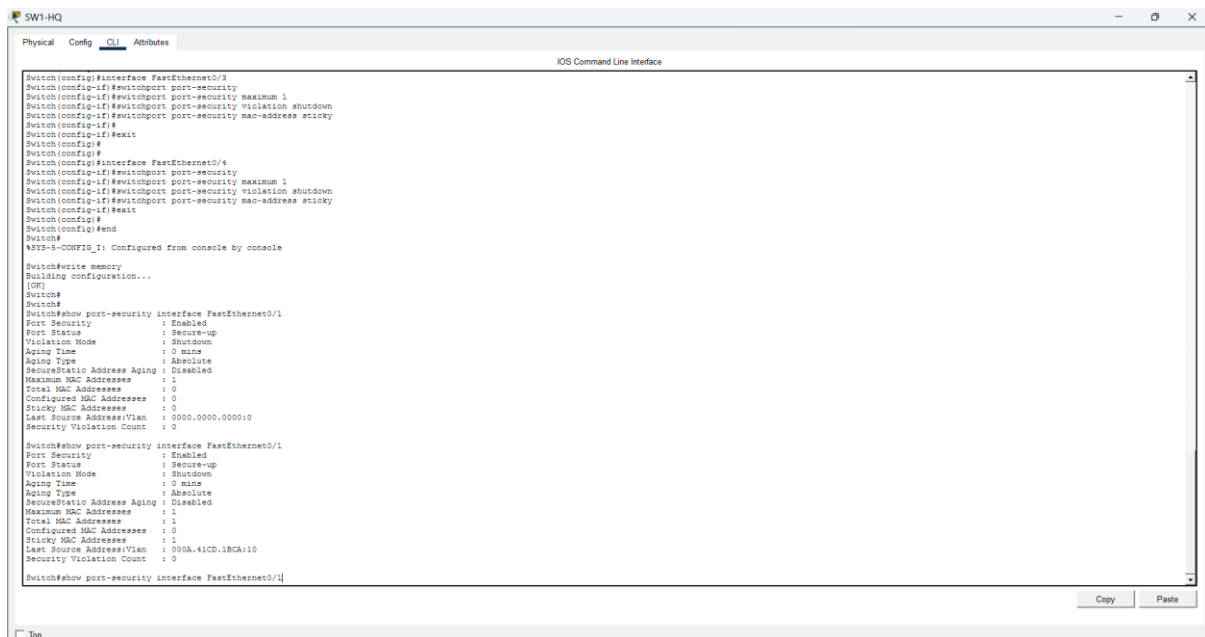
R2-Branch#enable
R2-Branch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2-Branch(config)#
R2-Branch(config)#ip access-list extended BLOCK-GUEST
R2-Branch(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
R2-Branch(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
R2-Branch(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
R2-Branch(config-ext-nacl)#permit ip any any
R2-Branch(config-ext-nacl)#exit
R2-Branch(config)#
R2-Branch(config)#interface GigabitEthernet0/0.50
R2-Branch(config-subif)#ip access-group BLOCK-GUEST in
R2-Branch(config-subif)#exit
R2-Branch(config)#
R2-Branch(config)#end
R2-Branch#
R2-Branch#show conf ig 1: Configured from console by console
R2-Branch#write memory
Building configuration...
[OK]
R2-Branch#
```



5.2 Port Security

Port security was enabled on all access ports of SW1-HQ to prevent unauthorized devices from connecting to the network. The sticky MAC address feature was used to automatically learn and lock the MAC address of the connected device. If a different device is plugged into any secured port, the port automatically shuts down.

```
interface FastEthernet0/1
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
```



```
SW1-HQ
Physical Config CLI Attributes
IOS Command Line Interface

Switch(config)#interface FastEthernet0/3
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface FastEthernet0/4
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#
MSTP-0-CONF10_1: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
Switch#show port-security interface FastEthernet0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#show port-security interface FastEthernet0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 1
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 000A.41CD.1BCA:10
Security Violation Count : 0

Switch#show port-security interface FastEthernet0/1
```

5.3 SSH Remote Management

SSH version 2 was configured on all four network devices (R1-HQ, R2-Branch, SW1-HQ, SW2-Branch) to enable secure encrypted remote management. Telnet was explicitly disabled. This ensures that all management traffic is encrypted and protected from eavesdropping.

```
ip domain-name company.local
username admin secret Str0ngP@ss123
crypto key generate rsa (1024 bits)
line vty 0 4
 login local
 transport input ssh
ip ssh version 2
```

```
R1-HQ
Physical Config CLI Attributes
IOS Command Line Interface

ALINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down
06:13:34: MOSPF-5-ADJCHG: Process 1, Hbr 192.168.50.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
ALINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
06:13:44: MOSPF-5-ADJCHG: Process 1, Hbr 192.168.50.1 on Serial0/0/0 from LOADING to FULL, Loading Done

R1-HQ#enable
R1-HQ#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1-HQ(config)#
R1-HQ(config)#ip domain-name company.local
R1-HQ(config)#username admin secret Str0ng$eal23
R1-HQ(config)#
R1-HQ(config)#crypto key generate rsa
The name for the keys will be: R1-HQ.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
R1-HQ(config)#
*Mar 1 6:13:52.259: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1-HQ(config)#
R1-HQ(config)#line vty 0 4
R1-HQ(config-line)#login local
R1-HQ(config-line)#transport input ssh
R1-HQ(config-line)#exit
R1-HQ(config)#
R1-HQ(config)#
R1-HQ(config)#ip ssh version 2
R1-HQ(config)#
R1-HQ(config)#end
R1-HQ#
*V15-5-CONFIG_1: Configured from console by console
R1-HQ#write memory
Building configuration...
[OK]
R1-HQ#
```

```
Top
SW1-HQ
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#ip domain-name company.local
Switch(config)#username admin secret Str0ng$eal23
Switch(config)#
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#hostname SW1-HQ
SW1-HQ(config)#
SW1-HQ(config)#crypto key generate rsa
The name for the keys will be: SW1-HQ.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
SW1-HQ(config)#
*Mar 1 6:13:52.259: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW1-HQ(config)#
SW1-HQ(config)#line vty 0 4
SW1-HQ(config-line)#login local
SW1-HQ(config-line)#transport input ssh
SW1-HQ(config-line)#exit
SW1-HQ(config)#
SW1-HQ(config)#
SW1-HQ(config)#ip ssh version 2
SW1-HQ(config)#
SW1-HQ(config)#end
SW1-HQ#
*V15-5-CONFIG_1: Configured from console by console
SW1-HQ#write memory
Building configuration...
[OK]
SW1-HQ#
```

```
Top
R2-Branch
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

R2-Branch#enable
R2-Branch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2-Branch(config)#
R2-Branch(config)#ip domain-name company.local
R2-Branch(config)#username admin secret Str0ng$eal23
R2-Branch(config)#
R2-Branch(config)#crypto key generate rsa
The name for the keys will be: R2-Branch.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
R2-Branch(config)#
*Mar 1 6:15:44.650: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2-Branch(config)#
R2-Branch(config)#
R2-Branch(config)#line vty 0 4
R2-Branch(config-line)#login local
R2-Branch(config-line)#transport input ssh
R2-Branch(config-line)#exit
R2-Branch(config)#
R2-Branch(config)#
R2-Branch(config)#ip ssh version 2
R2-Branch(config)#
R2-Branch(config)#end
R2-Branch#
*V15-5-CONFIG_1: Configured from console by console
R2-Branch#write memory
Building configuration...
[OK]
R2-Branch#
```



Security Feature	Purpose	Applied On
Extended ACL	Block Guest VLAN from HQ resources	R2-Branch Gi0/0.50
Port Security	Prevent unauthorized device connections	SW1-HQ Fa0/1 to Fa0/4
SSH v2	Encrypted remote management	R1, R2, SW1, SW2
Sticky MAC	Auto-lock authorized device MAC addresses	SW1-HQ access ports

6. Testing and Verification

6.1 Verification Results

All configurations were tested and verified using Cisco IOS show commands and end-to-end ping tests. The following table summarizes all test results.

Test	Command/Action	Result
VLANs active on SW1	show vlan brief	PASSED — VLANs 10,20,30 active
Trunk port working	show interfaces trunk	PASSED — Fa0/24 trunking
OSPF neighbor formed	show ip ospf neighbor on R1	PASSED — R2 state: FULL
Inter-VLAN routing	Ping PC1 to PC2 (VLAN10 to VLAN20)	PASSED — 3/4 replies
Cross-site connectivity	Ping PC1 (HQ) to PC4 (Branch)	PASSED — 3/4 replies
DHCP — HQ VLANs	PC1, PC2, PC3 set to DHCP mode	PASSED — IPs auto assigned
DHCP — Branch VLANs	PC4, PC5 set to DHCP mode	PASSED — IPs auto assigned
DNS resolution	Browser: hq.company.local	PASSED — server page loaded
ACL — Guest blocked	PC5 ping 192.168.10.2	PASSED — Request timed out
ACL — Staff allowed	PC4 ping 192.168.10.2	PASSED — Reply received
Port security	show port-security int Fa0/1	PASSED — Secure-up, max 1
SSH remote access	ssh -l admin 192.168.10.1 from PC1	PASSED — R1-HQ> prompt

6.2 OSPF Verification Output

The following output from R1-HQ confirms that OSPF neighbour adjacency was successfully formed with R2-Branch, with the neighbour state showing FULL — indicating complete route exchange between the two sites.

R1-HQ# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.50.1	0	FULL/ -	00:00:34	10.0.0.2	Serial0/0/0

```
R1-HQ
R1-HQ#enable
R1-HQ#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1-HQ(config)#
R1-HQ(config)#
R1-HQ(config)#interface Serial0/0/0
R1-HQ(config-if)#ip address 10.0.0.1 255.255.255.252
R1-HQ(config-if)#clock rate 64000
R1-HQ(config-if)#no shutdown
R1-HQ(config-if)#
%LINE-5-CHANGED: Interface Serial0/0/0, changed state to down
R1-HQ(config-if)#
R1-HQ(config-if)#exit
R1-HQ(config)#
R1-HQ(config)#router ospf 1
R1-HQ(config-router)#network 192.168.10.0 0.0.0.255 area 0
R1-HQ(config-router)#network 192.168.20.0 0.0.0.255 area 0
R1-HQ(config-router)#network 192.168.30.0 0.0.0.255 area 0
R1-HQ(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1-HQ(config-router)#exit
R1-HQ(config)#end
R1-HQ
%SYS-5-CONFIG_I: Configured from console by console
R1-HQ#write memory
Building configuration...
[OK]
R1-HQ#
%LINE-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
04:29:29: %OSPF-3-ADJCHG: Process 1, Nbr 192.168.50.1 on Serial0/0/0 from LOADING to FULL, Loading Done
R1-HQ#
R1-HQ#show ip ospf neighbor

Neighbor ID      Pri     State           Dead Time   Address      Interface
192.168.50.1      0      FULL/ -         00:00:34    10.0.0.2     Serial0/0/0
R1-HQ#
```

7. Lessons Learned

This project provided significant practical learning beyond what is typically covered in theoretical coursework. The following key lessons were gained through the hands-on implementation process:

7.1 Technical lessons

- The Cisco 3560 Layer 3 switch requires the 'switchport trunk encapsulation dot1q' command before enabling trunk mode — this is not required on the 2960 switch and was a critical troubleshooting discovery during the project.
- DHCP broadcast requests do not cross router boundaries by default. The ip helper-address command is essential for centralized DHCP to work across multiple VLANs and sites.
- The first ping in a new network typically times out due to ARP resolution — this is normal behaviour and does not indicate a network fault.
- OSPF convergence takes approximately 30 seconds after configuration before routes are exchanged and neighbour adjacency reaches FULL state.
- ACLs must be applied in the correct direction — inbound on the source interface — to function as intended.
- SSH key generation requires an ip domain-name to be configured first before the crypto key generate rsa command will execute.

7.2 Professional skills gained

- Systematic troubleshooting using show commands to isolate and identify configuration issues before applying fixes.
- The importance of planning IP addressing and VLAN design on paper before starting any device configuration.
- Documentation discipline — recording every configuration, test result, and verification output for professional reporting.
- Understanding how enterprise network security policies are implemented technically through ACLs and port security.
- Appreciation of how multiple network services (routing, DHCP, DNS, security) work together in an integrated enterprise environment.

8. Conclusion

This enterprise network simulation project successfully demonstrated the design, implementation, and verification of a fully functional two-site enterprise network. All planned features were implemented and verified including VLAN segmentation, inter-VLAN routing, OSPF dynamic routing, DHCP and DNS services, ACL-based security, port security, and SSH remote management.

The project provided invaluable hands-on experience with technologies and protocols that are used daily by network engineers in real enterprise environments. The troubleshooting challenges encountered during implementation — particularly around trunk port configuration and DHCP relay — deepened understanding of how these technologies work at a fundamental level.

The knowledge and skills gained through this project directly support my readiness for a network engineer or network security internship role, and demonstrate practical competency in Cisco IOS configuration, enterprise network design, and network security implementation.

KEY ACHIEVEMENT: The complete working simulation file (.pkt), topology diagram, and all configuration files are available on GitHub for review and verification by any network professional.